

- Q.27 What is electric drive and explain with the help of block diagram of an electric drive system.
- Q.28 Draw a circuit for a fan regulator using a TRIAC-DIAC pair.
- Q.29 How UPS can be obtained? Give its important applications
- Q.30 Explain the following
a) Voltage surge b) Brownouts
- Q.31 List five instructions for maintenance and care of lead acid battery.
- Q.32 Define cycloconverter. Describe basic principle.
- Q.33 Describe the working of parallel inverter.
- Q.34 Explain working principle of SCR with the help of V-I characteristics.
- Q.35 Explain single phase full wave DC converter drives.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write short technical notes on :
a) LASCR b) GTO
c) SUS d) PUT
d) IGBT
- Q.37 Explain PWM and frequency modulation schemes of chopper with the aid of waveforms. Which one is battery and why?
- Q.38 Explain the role of a DC-DC converter in a battery - powered devices and how it optimizes energy use from renewable sources like solar panels.

(20)

(4) 120953/030953/105853

No. of Printed Pages : 4 120953/030953/105853
Roll No.

5th Sem / Elect, Power Station Engg., Elect & Eltx Engg. Sub : Industrial Electronics & Control of Drives

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 SCR can be used as
a) Unidirectional switch
b) Bidirectional switch
c) Not used as a switch
d) All switches
- Q.2 A triac can be fired
a) Only by the positive half cycle of the supply
b) Only by the negative half cycle of the supply
c) Both by the positive and negative half cycles of the supply
d) Without any biasing
- Q.3 An inverter converts
a) AC into DC
b) DC into AC
c) AC into AC of different frequency
d) DC into variable DC
- Q.4 In Ward Leonard method of speed control
a) Fixed voltage variable frequency scheme is used
b) Field current control scheme is used
c) Armature voltage control is used
d) None of above

(1) 120953/030953/105853

- Q.5 In a Grid-tied solar PV system. What features allow excess electricity to be fed back into the grid?
- Power factor correction
 - Net metering
 - Battery charging
 - Anti islanding protection
- Q.6 Field control method is used in DC motors to get
- Speed below normal speed
 - Speed above normal speed
 - Both 'a' and 'b'
 - None of above
- Q.7 HVDC transmission systems generally used
- 6 pulse converter
 - 3 pulse converter
 - 12 pulse converters
 - Either 6 or 12 pulse converters
- Q.8 Cyclo converter can be
- Step up
 - Step down
 - Both step down and step up
 - Neither of above
- Q.9 UPS is never used in
- | | |
|--------------------|-----------------------|
| a) Computer | b) Communication link |
| c) Street lighting | d) All of above |
- Q.10 The V-I characteristics of DIAC lie in the
- 4th quadrant
 - 1st quadrant
 - The 1st & 3rd quadrant
 - The 2nd & 3rd quadrant

(2) 120953/030953/105853

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write two points for maintenance of battery.
- Q.12 Expand SMPS.
- Q.13 Enlist the types of choppers.
- Q.14 What do you mean by heat sink?
- Q.15 Draw V-I characteristics of SCR.
- Q.16 PLC stands for _____.
- Q.17 A UJT has _____ PN junction.
- Q.18 In half wave rectifier, one switching device is used. (True/False)
- Q.19 Define electric drive.
- Q.20 What do you mean by phase control?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working of step-down chopper with the help of circuit diagram.
- Q.22 Compare the different methods of triggering an SCR using data control.
- Q.23 Show the basic construction of a UJT, what is its intrinsic stand-off ratio.
- Q.24 Why three-phase rectifiers are preferred over single-phase rectifiers?
- Q.25 What is the need of conversion of AC supply to DC supply? Give some applications of thyristor converters.
- Q.26 Explain advantages and disadvantages of circulating current mode of dual converter.

(3) 120953/030953/105853